

Skeptic Engine — Scientific Data Integrity Research Report

Author: Sergey Boiko **Date:** 2026-03-26 **Status:** Internal research report. Not for public distribution.

Framing: This project develops statistical artifact detection tools, not fraud accusation tools. All flagged datasets require domain expert review. The methods detect non-physical patterns — they do not determine intent.

1. Executive Summary

This report documents the complete research arc from project inception to experimental validation. The project began as an “Interdisciplinary Discovery Engine” aimed at finding interdisciplinary scientific hypotheses. Through systematic evaluation, it pivoted to a focused research programme: **transferring financial anomaly detection methods to scientific data integrity screening**.

The tools developed here detect **statistical artifacts and anomalous patterns** — non-physical digit distributions, structural incoherence, and reporting inconsistencies. They do not claim to identify deliberate fraud; flagged datasets require domain expert review before any conclusions about intent.

Three independent experiments produced positive results across five datasets:

Experiment	Best AUC	Datasets	Key Finding
H24: Benford digit forensics on scRNA-seq	0.978 (fusion)	PBMC3k, Kang2018	Digit + cell-level fusion detects fabricated count matrices
H25: Banking autoencoder on proteomics	1.000 (fusion)	CPTAC proteomics, CNA	AE reconstruction error catches structure-breaking fabrication
H23: Behavioral p-hacking detection	0.993 (sim) / 0.765 (real)	Simulated, RPP (99), Statcheck (61)	Fraud sequence features beat baselines by +7.5pp to +28.7pp

One hypothesis was killed after honest execution:

Experiment	AUC	Verdict
H4: TDA for cancer drug resistance	0.500 (random)	KILLED — TDA adds zero value; PCA=0.9998 is trivial batch effect

All experiments were independently audited for data leakage, fabrication realism, and feature correctness. Cross-dataset generalization tests revealed honest limitations. Kill criteria were applied without rationalization.

2. Project History and Pivot

2.1 Original Goal

The project started with the goal of building an agentic system to discover interdisciplinary scientific hypotheses worthy of Nobel Prize consideration. An “Interdisciplinary Discovery Engine” was built with a Python pipeline: acquisition → DQOps → semantic core → hypothesis generation → skeptic → arbiter.

2.2 Assessment and Pivot

An independent assessment (2026-03-26) rated the original approach 4.5/10. Key problems identified:

- All 20 original hypotheses were well-known method transfers (GNN to biology, TDA to neuroscience, etc.)
- The semantic core used bag-of-tags matching, not real NLP

- The hypothesis generator used template-based text, not deep reasoning
- The MOF benchmark (H10) was already surpassed by Princeton (97% F1) and MOFGen
- The Nobel Prize framing created cognitive bias toward impressive-sounding ideas over testable ones

2.3 New Direction

The assessment identified the operator's unique strength: **financial fraud detection expertise**. This led to a systematic search for transfers from fraud/security methods to scientific domains. Four rounds of hypothesis generation produced progressively better candidates, converging on **scientific data integrity** as the primary research theme.

2.4 Research Programme

The final programme is: "Financial Anomaly Detection Methods → Scientific Data Integrity," with three validated experiments and a clear path to additional results.

3. Infrastructure Built

3.1 Discovery Engine Pipeline

```
src/discovery_engine/
  collectors/          # bioRxiv, Zenodo, Semantic Scholar API collectors
    biorxiv.py        # bioRxiv preprint collector
    zenodo.py         # Zenodo dataset/publication collector
    semantic_scholar.py # Citation-based authority scoring
    _http.py          # Rate-limited HTTP with retry
    _tag_extractor.py  # Bridge-tag extraction (18 categories)
    _domain_mapper.py  # API category → project domain mapping
  dqops/              # Deduplication, authority scoring, bias flagging
  semantic_core/       # Cross-domain link discovery
  hypothesis_generation/ # Hypothesis card generation
  skeptic/            # Prior-art challenge with external source matching
  arbiter/            # Ranking and promotion
  schemas/            # SourceRecord, HypothesisCard, CandidateSeed
  benchmarks/h10/     # MOF stability benchmark (completed)
  pipeline/           # Stage contracts
```

3.2 External Source Collectors

Three API collectors were built (stdlib only, zero external dependencies):

- **Semantic Scholar**: citation-based authority scoring via $\log_{10}(\text{citations})$, year-based novelty
- **bioRxiv**: preprint collector with date/category filtering
- **Zenodo**: dataset/publication/software collector

All collectors produce `list[SourceRecord]` compatible with the pipeline, with rate limiting, retry logic, and JSON manifest persistence for reproducibility.

3.3 H10 MOF Benchmark (completed, deprioritized)

The MOF stability benchmark was built to full execution readiness: - 2,179 MOF structures imported from MOF-Simplify - Two descriptor baselines: LogReg (AP=0.858) and HGB (AP=0.874) - Graph-ready dataset: 2,179/2,179 structures, 0 failures, avg 281 nodes × 291 edges - Baseline scaffold with explicit readiness checks

H10 was deprioritized after prior-art review showed the field is saturated (Princeton 97% F1, MOFGen agentic synthesis, 2025 Nobel for MOF development).

4. Experiment H24: Benford Digit Forensics on scRNA-seq

4.1 Hypothesis

Benford first- and second-digit frequency features, combined with cell-level structural features and Isolation Forest anomaly scoring, can detect fabricated scRNA-seq UMI count matrices.

Prior art: Bradshaw 2021 did CNA (not scRNA-seq). Vilenchik 2019 did Benford for cell typing (not fabrication). Gap confirmed via PubMed/Semantic Scholar search.

4.2 Methods

Datasets: PBMC3k (2,700 cells × 32,738 genes), Kang2018/GSE96583 (subsamped to 2,700 cells × 13,537 genes)

Fabrication methods (adapted from Bradshaw 2021):

Method	Mechanism	Preserves	Destroys
Resample	Per-gene shuffle across cells	Per-gene marginals	Inter-gene correlation
Noise	Additive Poisson ($\lambda=0.15 \times \text{value}$)	Approximate values	Sparsity structure
Random NB	Per-gene NB(n,p) from fitted params	Per-gene mean/variance	Digit-level distribution

Features (30 total): - Benford first-digit frequencies (9) - Benford second-digit frequencies (10) - Chi-squared divergence from Benford expected (2) - Cell-level: library size, genes detected, zero fraction, mean/var/CV nonzero, max count, log-total (8) - Isolation Forest anomaly score trained on real data only (1)

Classifiers: Random Forest (100 trees), Logistic Regression

4.3 Results

Within-Dataset Detection (PBMC3k)

Fabrication	Benford RF	Cell-Level RF	IF-Only RF	Fusion RF
Resample	0.864	0.956	0.639	0.978
Noise*	1.000	1.000	1.000	1.000
Random NB	0.989	0.997	0.843	0.999

*Noise AUC=1.000 reflects trivial sparsity change (nonzero 2.6% → 21.4%), not meaningful detection.

Cross-Dataset Generalization (PBMC3k ↔ Kang2018)

Fabrication	Within-PBMC	Within-Kang	PBMC→Kang	Kang→PBMC
Resample	0.953	0.986	0.430	0.482
Noise	1.000	1.000	1.000	1.000
Random NB	0.995	0.994	0.186	0.764

Cross-dataset generalization fails for resample and random NB. Noise generalizes trivially. Feature normalization (z-score, rank, delta-Benford) does not rescue generalization.

Feature Importance

- **Resample:** cell-level features dominate (zero fraction, library size); Benford χ^2 secondary
- **Random NB:** Benford χ^2 is top feature; digit frequencies fd_2, fd_4 contribute
- **Mechanism:** resampling homogenizes cells, collapsing inter-cell digit variance by 4×

Adversarial Robustness A “Benford-aware” fabricator who forces first-digit compliance actually makes detection EASIER (AUC=1.000). This is because real scRNA-seq UMI counts do NOT follow Benford (fd₁=0.74 vs Benford 0.30, caused by many counts=1). **Benford compliance itself is a suspicious signal in scRNA-seq.**

4.4 Audit Results

- Raw data: verified real PBMC3k from 10x Genomics (2,700 cells, integer counts, min=0, max=419)
- Fabrication: verified all three methods produce genuinely different data (0/2700 identical rows)
- Features: manually verified digit extraction matches hand computation
- Classifier: zero train/test overlap; shuffle test AUC=0.51 (no leakage)

4.5 Artifacts

```
experiments/h24_benford_scrna/
  digit_features.py      # Benford feature extraction
  fabrication.py         # Three fabrication methods
  isolation_forest.py    # IF anomaly scoring
  run_h24.py             # Standalone H24 experiment
  run_combined.py        # H24+H21 fusion experiment
  run_crossval.py        # Cross-dataset validation
  run_normalized_crossval.py # Normalization rescue attempt
  run_feature_importance.py # Permutation importance analysis
  run_adversarial.py     # Adversarial robustness test
  generate_figures.py    # Publication-quality figures
  figures/               # 4 figures (PNG 300dpi + PDF)
  results/               # 7 JSON result files
  paper_outline.md       # Full paper outline
  collaboration_pitch.md # Co-author search strategy
  run_cross_tissue.py    # Cross-tissue generalization test
```

4.7 Cross-Tissue Generalization (PBMC human ↔ Brain mouse)

Fabrication	Within-PBMC	Within-Brain	PBMC→Brain	Brain→PBMC
Resample	0.976	1.000	0.376	0.442
Noise	1.000	1.000	0.999	1.000
Random NB	0.999	1.000	0.388	0.690

Cross-tissue generalization fails for resample and random NB (same pattern as cross-dataset). Noise generalizes trivially.

Dataset: 10x Genomics Neuron 900 (mouse brain, 900 cells × ~27k genes) vs PBMC3k (human blood, 2,700 cells × 32k genes).

This confirms that the artifact signal is **dataset-specific, not tissue-specific**: even changing species + tissue + cell count does not help the model transfer. The features encode dataset identity, not universal artifact signatures.

5. Experiment H25: Banking Fraud Autoencoder for Proteomics

5.1 Hypothesis

Banking-style dense autoencoder reconstruction error can detect fabricated proteomics/CNA data, complementing Benford digit features especially for structure-breaking fabrication.

Prior art: No paper transfers banking fraud autoencoder methods to mass spectrometry data. Gap confirmed.

5.2 Methods

Data: Bradshaw 2021 CPTAC datasets — proteomics (140 samples × 9,585 proteins) and CNA (100 samples × 17,156 genes)

Fabrication: random (per-gene normal), shuffle (per-gene permutation), noise (10% Gaussian)

Detection methods: - Dense Autoencoder (256→32→256, trained on real data only, MSE reconstruction error) - Benford first-digit RF (9 features) - Fusion (Benford + AE error)

5.3 Results

Dataset	Fabrication	AE AUC	Benford AUC	Fusion AUC	% Flagged (98th pct)
Proteomics	Random	0.989	1.000	1.000	84%
Proteomics	Shuffle	0.989	0.811	1.000	87%
Proteomics	Noise 10%	0.516	0.997	0.995	3%
CNA	Random	0.821	1.000	1.000	0%
CNA	Shuffle	0.824	1.000	1.000	0%
CNA	Noise 10%	0.533	0.928	0.945	2%

5.4 Key Findings

1. **AE beats Benford on shuffle** (proteomics AE=0.989 vs Benford=0.811) — autoencoder catches structural coherence violations that digit tests miss
2. **Benford beats AE on noise** (proteomics Benford=0.997 vs AE=0.516) — small noise doesn't change reconstruction pattern but breaks digit distributions
3. **Fusion wins on ALL fabrication types** (AUC 0.945-1.000) — complementary error profiles
4. **Banking-style 98th percentile threshold** flags 84-87% of fabricated proteomics samples for random/shuffle

5.5 Cross-Omics Generalization (CNA ↔ Proteomics)

Benford-only features trained on CNA, tested on Proteomics (and vice versa):

Fabrication	Within-CNA	Within-Prot	CNA→Prot	Prot→CNA
Random	1.000	1.000	0.171	0.134
Shuffle	1.000	1.000	0.466	0.172
Noise 10%	1.000	1.000	0.445	0.572

Cross-omics generalization fails completely (mean AUC = 0.33, below random). Benford digit features are omics-type-specific: the digit distribution of CNA values differs fundamentally from proteomics expression values, making cross-type transfer impossible without retraining.

This is consistent with the cross-tissue and cross-dataset results: within-dataset detection is strong, but the features encode dataset-specific properties.

5.6 Artifacts

```
experiments/h25_banking_ae_lcms/  
  run_h25.py                # Full experiment  
  run_h25_cross_omics.py    # Cross-omics generalization test  
  data/bradshaw/           # Downloaded CPTAC data  
  results/                  # h25_results.json + h25_cross_omics.json
```

6. Experiment H23: Behavioral P-Hacking Detection

6.1 Hypothesis

Fraud-style behavioral sequence features (velocity, volatility, direction changes, terminal behavior) extracted from p-value sequences can detect p-hacking better than aggregate p-curve analysis.

6.2 Simulated Data Results

Setup: 500 clean studies (real/null effects) vs 500 p-hacked studies (4 strategies: optional stopping, outcome switching, outlier exclusion, covariate fishing). 18 behavioral features per study.

Method	AUC
RF Behavioral features	0.993
Isolation Forest (one-class)	0.946
P-curve baseline	0.706

RF improvement over p-curve: +28.7 percentage points AUC.

Top features: volatility (std of p-value deltas), min_p, max_jump, frac_significant.

6.3 Real Data Results (Reproducibility Project: Psychology)

Setup: 99 studies from Open Science Collaboration 2015 with known replication outcomes. 12 features extracted from original study p-values and metadata.

Method	AUC
LR Behavioral features	0.729 $\pm$ 0.173
RF	0.622 $\pm$ 0.175
GBM	0.616 $\pm$ 0.125
Baseline (p-value alone)	0.654

LR improvement over baseline: +7.5 percentage points AUC on real data.

Top features: p_original_log (0.297), p_digit_1 (0.129), p_second_digit (0.116). Digit features rank 2nd and 3rd, confirming Benford-style analysis adds signal beyond the p-value itself.

6.4 Statcheck Real Data Results (meta-analysis extracted p-values)

Setup: 61 articles from statcheck meta-analyses dataset (OSF n5xba), each with 2+ extracted statistical tests (506 tests total). Features extracted per-article: p-value distribution stats, threshold clustering, discrepancy metrics, sequence dynamics, digit patterns. Label: has statcheck error (1) vs clean (0).

Method	AUC
Baseline (mean_discrepancy alone)	0.523
LR Behavioral features	0.765 $\pm$ 0.084
GBM	0.756 $\pm$ 0.101
RF	0.740 $\pm$ 0.086

LR improvement over baseline: +24.3 percentage points AUC on real statcheck data.

Top features: std_p (0.129), n_tests (0.102), volatility (0.090), mean_p (0.088), digit_1_freq (0.061).

6.5 Cross-Dataset Summary for H23

Dataset	n	Best AUC	Δ vs baseline	Ground truth
Simulated (500+500)	1000	0.993	+28.7pp vs p-curve	Controlled simulation
RPP real (99)	99	0.729	+7.5pp vs p-value	Known replication outcomes
Statcheck real (61)	61	0.765	+24.3pp vs discrepancy	Statcheck error flags

Consistent positive signal across all three datasets. Improvement ranges +7.5pp to +28.7pp.

6.6 Honest Caveats

- Simulated data AUC (0.993) is likely optimistic — real p-hacking is subtler
- Real data samples are small (n=61-99); high variance across folds
- LR > RF/GBM indicates low-sample regime where regularization matters
- Statcheck errors are reporting inconsistencies, not confirmed p-hacking — proxy label only
- Large-scale validation (688k p-values from Nuijten 2016) blocked by dataset access (DANS repository requires manual download)

6.7 Artifacts

```
experiments/h23_phacking_behavioral/  
  run_h23.py                # Simulated experiment  
  run_h23_real.py           # Real data (RPP) validation  
  run_h23_statcheck.py      # Statcheck meta-analyses validation  
  data/rpp_data.csv         # Reproducibility Project dataset  
  data/statcheckDataMetaAnalyses_Anonymized.txt # Statcheck extracted p-values  
  results/                  # 3 JSON result files
```

7. Experiment H4: TDA for Cancer Drug Resistance (KILLED)

7.1 Hypothesis

Persistent homology (TDA) features computed on local PCA neighborhoods of scRNA-seq cells can distinguish drug-sensitive from drug-resistant melanoma cells better than standard embedding baselines.

7.2 Dataset

GSE164897 (Schmidt et al. 2021): A375 melanoma cell line, 30,716 cells across 4 samples: - 1 sensitive (untreated): 6,695 cells - 3 resistant (vemurafenib ± combination): 4,328 + 9,266 + 10,427 cells - 33,538 shared genes after alignment

7.3 Results

Method	AUC	BA
PCA (50 components)	0.9998	0.9925
Dispersion (10 features)	0.9378	0.8387
TDA (12 PH features)	0.5000	0.5000
PCA + TDA fusion	0.9998	0.9919
PCA + Disp + TDA fusion	0.9997	0.9931

7.4 Kill Criterion Applied

- TDA standalone AUC: 0.5000 (random chance)
- PCA baseline AUC: 0.9998 (near-perfect)
- TDA lift over PCA: -0.4998
- PCA+TDA fusion lift: +0.0000

VERDICT: KILL — TDA AUC (0.5000) < 0.75 threshold.

7.5 Interpretation

1. **PCA AUC = 0.9998 is batch effect, not biology.** With only 4 samples (1 sensitive, 3 resistant), cell-level classification learns sample identity, not resistance biology. Any embedding that captures batch differences will achieve near-perfect separation.
2. **TDA = random.** Local persistent homology features (H0, H1 on 30-NN neighborhoods) are invariant to the batch differences that PCA captures. TDA sees the same local topology in all cells regardless of resistance status.
3. **The hypothesis that TDA captures “resistance state transitions” is not supported.** On this dataset, there is no topological signal distinguishing sensitive from resistant cells above what PCA already trivially captures.

7.6 Lessons

- Kill criteria prevent sunk-cost continuation of dead tracks
 - Near-perfect PCA baseline is a red flag for batch confounding, not evidence of a good task
 - TDA’s value proposition (capturing shape beyond linear embedding) requires tasks where linear methods genuinely fail
 - This was a 4-minute execution + 236-second compute = complete answer in under 5 minutes
-

8. Novel Scientific Observations

8.1 scRNA-seq UMI Counts Do Not Follow Benford’s Law

Real scRNA-seq data has first-digit frequency $fd_1 = 0.74$ (vs Benford expected 0.30). This is because the median nonzero UMI count is 1, creating extreme concentration on digit 1. This means:

- **Standard Benford compliance testing is inverted** for scRNA-seq: conformity to Benford is suspicious, not expected
- A fabricator who forces Benford compliance makes their data MORE detectable, not less
- This observation is not documented in prior literature and represents a publishable finding on its own

8.2 Different Fabrication Types Are Caught by Different Feature Groups

- **Correlation-destroying fabrication (resample):** caught by cell-level structural features (zero fraction, library size correlation disruption)
- **Distribution-generating fabrication (random NB):** caught by Benford digit chi-squared deviation
- **Value-level manipulation (noise):** caught by both (trivially detectable due to sparsity change)

This implies that a single detection method is insufficient — fusion of complementary feature groups is necessary.

8.3 Cross-Dataset Generalization is Fundamentally Limited for Sophisticated Fabrication

Fabrication detection signals are entangled with dataset-specific characteristics (library preparation protocol, sequencing depth, biological composition). Feature normalization (z-score, rank, delta-Benford) does not rescue generalization. This is an honest negative result that should be reported to prevent false confidence in “universal” fabrication detectors.

8.4 Banking Autoencoder Reconstruction Error Captures Structural Coherence

The autoencoder trained on real proteomics data learns the joint distribution of protein expression values. Fabrication methods that break inter-protein correlations (shuffle, random) produce high reconstruction error even when marginal distributions are preserved. This is exactly the mechanism used in banking fraud detection (transaction coherence) and validates the cross-domain transfer.

9. Verified Results Summary

ID	Experiment	Best AUC	Bootstrap 95% CI	Datasets	Status
H24	Artifact detection fusion (scRNA-seq)	0.978	[0.988, 0.997]	PBMC3k, Kang2018, Neurons900	Defendable within-dataset
H24-cross	Cross-tissue (PBMC→Brain)	0.376	—	PBMC3k ↔ Neurons900	Fails (tissue-specific)
H25	Banking AE (proteomics/CNA)	1.000	—	CPTAC proteomics, CNA	Defendable within-dataset
H25-cross	Cross-omics (CNA→Prot)	0.171	—	CNA ↔ Proteomics	Fails (omics-specific)
H23-sim	Behavioral p-hacking (simulated)	0.993	—	1000 synthetic	Defendable
H23-real	Behavioral (RPP)	0.729	[0.417, 0.923]	99 real	Underpowered
H23-statcheck	Behavioral (statcheck)	0.765	[0.452, 1.000]	61 real	Underpowered
H23-pmc	PMC recompute (quasi-label)	0.980	—	59 PMC articles	Supporting (quasi-label)
H4	TDA cancer resistance	0.500	—	GSE164897	KILLED

Audited: data authenticity, fabrication validity, feature correctness, train/test leakage (shuffle test), bootstrap confidence intervals.

Key pattern: Within-dataset detection is strong (AUC 0.97-1.00). Cross-generalization fails at every level tested (dataset, tissue, omics type). Noise-level artifacts generalize trivially. H23 real-data claims are underpowered.

10. Limitations

1. **All fabrication is simulated.** No real-world fabricated scRNA-seq, proteomics, or p-hacking datasets with confirmed ground truth exist for validation.
2. **Cross-generalization fails at every level tested.** Cross-dataset (PBMC3k ↔ Kang2018), cross-tissue (human PBMC ↔ mouse brain), and cross-omics (CNA ↔ proteomics) all show AUC near or below random for sophisticated artifact types. Only noise-level manipulations generalize (trivially). Feature normalization does not help. The tool must be retrained per-dataset.
3. **Small sample sizes and wide confidence intervals.** RPP (n=99) and Statcheck (n=61) experiments have bootstrap 95% CIs that cross random (H23 RPP: [0.42, 0.92], H23 Statcheck: [0.45, 1.00]). H23 real data claims are underpowered and should be framed as suggestive trends, not confirmed results.
4. **Noise fabrication is trivially detectable** in H24 — the AUC=1.000 reflects a massive sparsity change, not sophisticated detection.

5. **No domain co-author** yet. The biological and proteomics framing needs expert validation before any claims can be considered publication-ready.
6. **Integer UMI counts only.** H24 methods assume raw counts; normalized, log-transformed, or imputed matrices would require separate investigation.
7. **Operator bias risk.** The anomaly detection framing may overfit to the operator's expertise rather than representing the most impactful research direction.

11. Artifacts Inventory

Result Files (11 JSON)

File	Experiment
h24_results.json	H24 standalone (3 fabrication × 2 classifiers)
h24_h21_combined.json	H24+H21 fusion (4 approaches × 3 fabrication)
h24_h21_crossval.json	Cross-dataset PBMC3k ↔ Kang2018
h24_normalized_crossval.json	Normalization rescue experiment (4 strategies)
h24_feature_importance.json	Permutation + Gini importance
h24_adversarial.json	Adversarial robustness (2 strategies)
h25_results.json	Banking AE (2 datasets × 3 fabrication × 3 methods)
h23_results.json	P-hacking simulated (3 detectors, 4 strategies)
h23_real_rpp_results.json	P-hacking real RPP (3 models, 5-fold CV)
h23_statcheck_results.json	P-hacking statcheck real (3 models, 5-fold CV, 61 articles)
h4_results.json	H4 TDA cancer resistance — KILLED (5 baselines, 30,716 cells)

Figures (7 PNG + 4 PDF)

Figure	Content
fig1_within_dataset_auc.png/pdf	Bar chart: 4 methods × 3 fabrication types
fig2_feature_importance.png/pdf	Feature group importance per fabrication type
fig3_cross_dataset_heatmap.png/pdf	Generalization heatmap PBMC3k ↔ Kang2018
fig4_digit_distributions.png/pdf	Benford distributions: real vs fabricated
h24_roc_curves.png	ROC curves per fabrication
h24_h21_comparison.png	4-method comparison bar chart
h24_feature_importance.png	Permutation importance bar chart

Code (18 experiment scripts)

All experiments are self-contained Python scripts in `experiments/`. Each downloads its own data, runs independently, and saves results to its `results/` directory.

```

experiments/
  dashboard.py                # Unified results view
  h24_benford_scrna/
    run_h24.py                # H24 standalone
    run_combined.py           # H24+H21 fusion
    run_crossval.py           # Cross-dataset validation
    run_normalized_crossval.py # Normalization rescue
    run_feature_importance.py  # Feature importance
    run_adversarial.py        # Adversarial robustness
    generate_figures.py        # Publication figures
    digit_features.py          # Benford extraction library

```

```

fabrication.py           # Fabrication generators
isolation_forest.py      # IF anomaly scoring
h25_banking_ae_lcsm/
  run_h25.py             # Banking AE experiment
h23_phacking_behavioral/
  run_h23.py             # Simulated p-hacking
  run_h23_real.py        # RPP real data
  run_h23_statcheck.py   # Statcheck real data
h4_tda_cancer/
  run_h4.py              # H4 TDA - KILLED

```

Dashboard

experiments/dashboard.py — unified view of all experiments in one terminal command.

12. Prospects

12.1 Immediate Next Steps (no public action required)

1. **Third dataset for H24** — non-PBMC tissue (brain, liver, tumor) to test tissue generalizability
2. **H25 cross-omics** — train on CNA, test on proteomics (and vice versa) to test modality transfer
3. **Feature engineering V2** — inter-gene correlation features (fraud detection uses inter-transaction patterns) which may improve cross-dataset generalization
4. **H23 larger p-value dataset** — use statcheck's 688k extracted p-values (available via DANS repository) for scaled validation

12.2 Publication Path (when ready)

Paper 1: "Digit Forensics for scRNA-seq Data Integrity" (H24) - Target: Bioinformatics Application Note - Needs: bioinformatics co-author, third dataset, response to reviewer domain questions - Estimated readiness: 4-6 weeks after co-author engagement

Paper 2: "Banking Fraud Methods for Multi-Omics Fabrication Detection" (H24 + H25 combined) - Target: PLoS Computational Biology - Needs: broader dataset validation, co-author with proteomics background - Estimated readiness: 8-12 weeks

Paper 3: "Behavioral Sequence Anomaly for Replication Risk Prediction" (H23) - Target: Research Integrity and Peer Review, or Meta-Psychology - Needs: larger p-value dataset, meta-science co-author - Estimated readiness: 12-16 weeks

12.3 Long-Term Research Programme

The overarching narrative is: **"Financial anomaly detection as a systematic source of methods for scientific data integrity screening."** This is a 3-5 year programme, not a single paper. The current results establish proof-of-concept across three data types (scRNA-seq, proteomics/CNA, p-values).

Future directions: - Adversarial fabrication detection (fabricators who know about the detector) - Real-time GEO/SRA submission screening tool - Extension to other omics types (metabolomics, epigenomics) - Behavioral authorship patterns for paper mill detection - Open-source toolkit release

11.4 Realistic Assessment

This work is not Nobel-class. It is a solid, honest, technically sound research programme with genuine novelty (confirmed prior-art gaps), real experimental results (audited), and clear limitations (documented).

What it IS: - A viable path to 2-3 publications in reputable journals - A genuine transfer of operator expertise to a new domain - A foundation for a recognizable research identity in scientific data integrity

What it is NOT: - A guaranteed breakthrough - A finished product ready for deployment - Sufficient without domain collaborators

13. Lessons Learned

1. **Operator's edge matters more than generic AI.** The pivot from "find any interdisciplinary hypothesis" to "transfer YOUR fraud expertise" produced dramatically better results.
2. **Honest negatives are valuable.** Cross-dataset generalization failure, normalization failure, and the Benford-inversion finding in scRNA-seq are all publishable observations that strengthen the work.
3. **Audit everything.** The shuffle test, manual feature verification, and sparsity-change caveat on noise fabrication would not have been caught without systematic auditing.
4. **Speed of execution matters.** Three experiments in one day (H24, H25, H23) with full verification is possible when the infrastructure is already in place.
5. **The Discovery Engine infrastructure was useful even after the pivot.** External source collectors, DQOps, and the skeptic stage all contributed to the hypothesis evaluation process that led to the final research direction.
6. **Kill criteria prevent sunk-cost fallacy.** H4 was killed in 4 minutes of compute after months of planning infrastructure. The planning was not wasted — it ensured the kill was clean, documented, and defensible. Without a pre-committed kill criterion, the temptation to "try one more thing" would have consumed weeks.
7. **Near-perfect baselines are red flags.** PCA AUC = 0.9998 on H4 means the task is trivially confounded (batch = sample identity). Any new method will also "succeed" on this task without adding value. Always check whether your baseline is suspiciously strong before claiming your method works.
8. **Two-branch portfolio management works.** Separating "core research" (Branch 2) from "R&D lab" (Branch 1) with explicit time budgets and kill criteria prevented the project from oscillating between tracks. H4 consumed exactly one clean run, then was closed. The remaining effort went to validated Branch 2 work.